(11) What is an addressing mode? Explain various types of addressing mode with example.

- An addressing mode is a method used by the cpu to specify the address of an operand required for an instruction.

→ different ways in which the location of an operand is specified in an instruction.

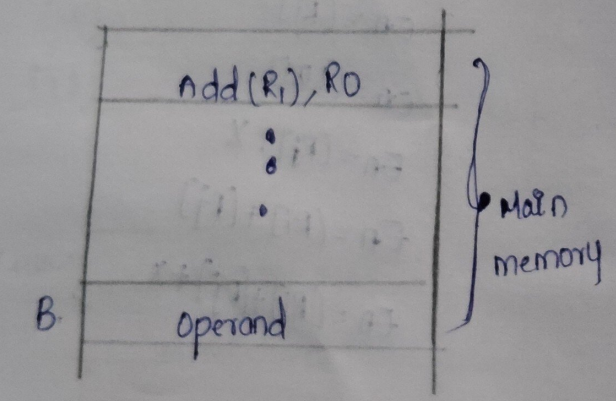
Generic addressing modes:

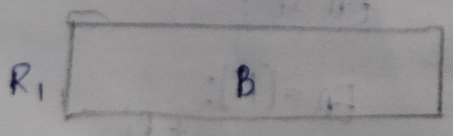
Name	Assembler syntax	Addressing function
Immediate	value	operand = value
Register	Ri	EA = Ri
Absolute (Direct)	LOC	EA = LOC
Indirect	(Ri)	EA = [Ri]
	(LOC)	EA = [LOC]
Index	X(Ri)	EA = [Ri] + X
Base with index	(Ri, Rj)	EA = [Ri] + [Rj]
Base with index and offset	X(Ri, Rj)	EA = [Ri] + [Rj] + X
Relative	X(PC)	EA = [PC] + X
Auto increment	(Ri)+	EA = [Ri]; Increment Ri
Auto decrement	-(Ri)	Decrement Ri; EA = [Ri]

Register mode:-
→ The operand is the contents of a register
→ The name (or the register is given in the instruction
→ For example.
move R_1, R_2 : copy content of register R_1 into register R_2

Absolute (Direct) Mode:-
* The operand is in a memory-location.
* For example, the instruction,
Move LOC, R_2 ; copy content of memory-location LOC into register R_2

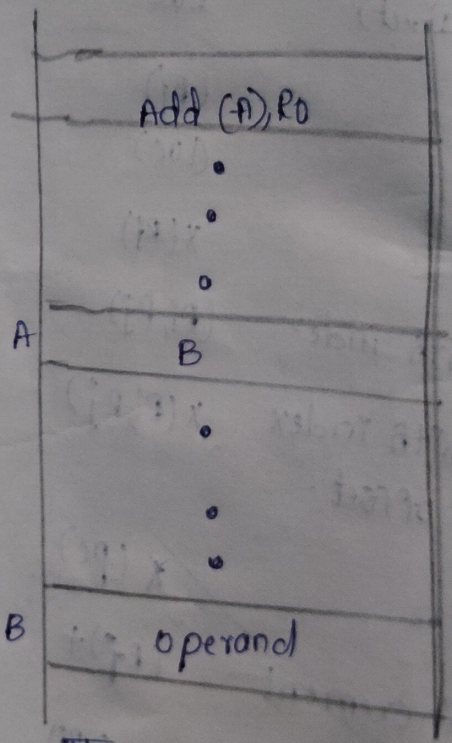
Indirect mode:-
* The EA of the operand is the contents of a register.
* The register that contains the address of an operand is called a pointer.





Register.

Through a general-purpose register



Through a memory location.

(12) Explain various methods for handling multiple interrupts requests raised by multiple devices.

In a computer system, several I/O devices such as keyboard, mouse, printer, disk and network devices may request the attention of the cpu simultaneously.

Methods for Handling Multiple interrupt Requests

Polling:-

→ Information needed to determine whether a device is requesting an interrupt is available in its status-register.

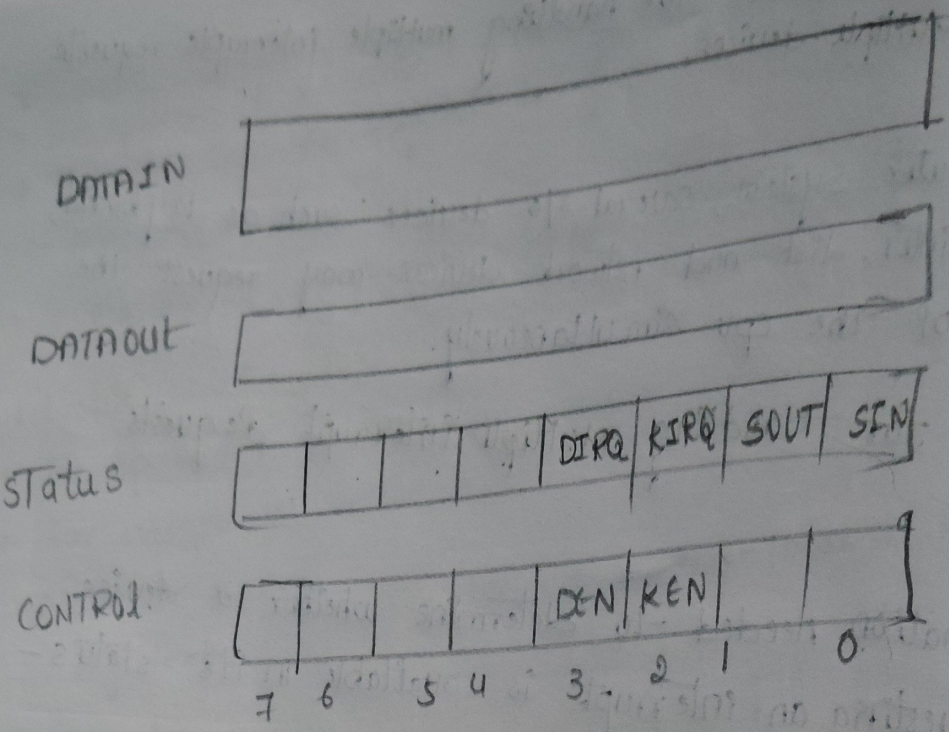
working:-

1. cpu receives an interrupt signal
2. cpu checks device 1
3. If not requesting, checks device 2.

→ when a device raises an interrupt-request, it sets IRQ bit to 1 in its status-register.

→ KIRQ and DIRQ are the interrupt-request bits for keyboard & display.

→ simplest way to identify interrupting device is to have ISR poll all I/O devices connected to bus.



9. Daisy chaining.

In this method, interrupt lines are connected in series. The interrupt request passes from one device to another based on priority.

Working:-

→ All devices share a common interrupt line
→ cpu sends an intterrupt acknowledge signal

priority:-

→ Device closest to cpu has highest priority
→ Last device has lowest priority.

Example:-

→ Disk → keyboard → printer
→ Disk interrupt serviced first

Advantages:-

→ faster than polling
→ simple hard ware structure

Disadvantages:-

→ fixed priority.

3. parallel priority interrupt

In this method each device has a separate interrupt request line.

Working:-

→ Devices send interrupt requests simultaneously

→ cpu services that interrupt first.

Example:-

Interrupts from keyboard, mouse and disk.

Advantages:-

→ very fast interrupt handling

→ NO need for polling

Disadvantages:-

* complex hardware

* costly implementation.

(H) What is cache memory? Analyze three mapping functions of cache memo

Ans: Processor is much faster than the main memory.

→ As a result, the processor has to spend much of its time waiting while instructions and data are being fetched from the main memory.

→ This serves as a major obstacle towards achieving good performance.

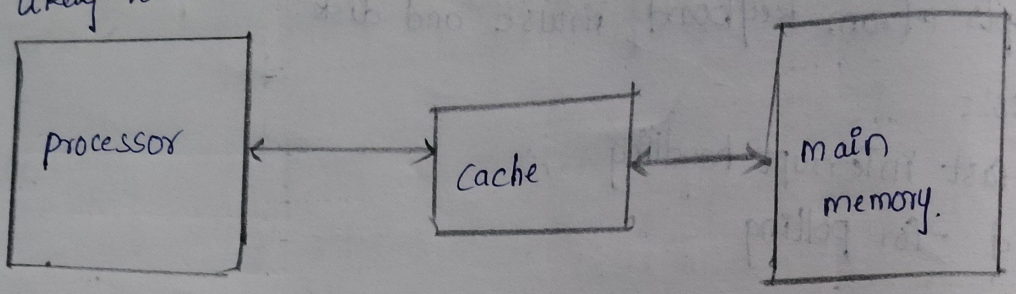
→ cache memory is based on the property of computer program known as "locality of reference".

Temporal locality of reference:↓

Recently executed instruction is likely to be executed again very soon.

Spatial locality of reference:↓

Instruction with addresses close to a recently instruction are likely to be executed soon.



→ which blocks in the main memory are in the cache determained by a mapping function.

Mapping functions:↓

(1) Direct Mapping

(2) Associative mapping

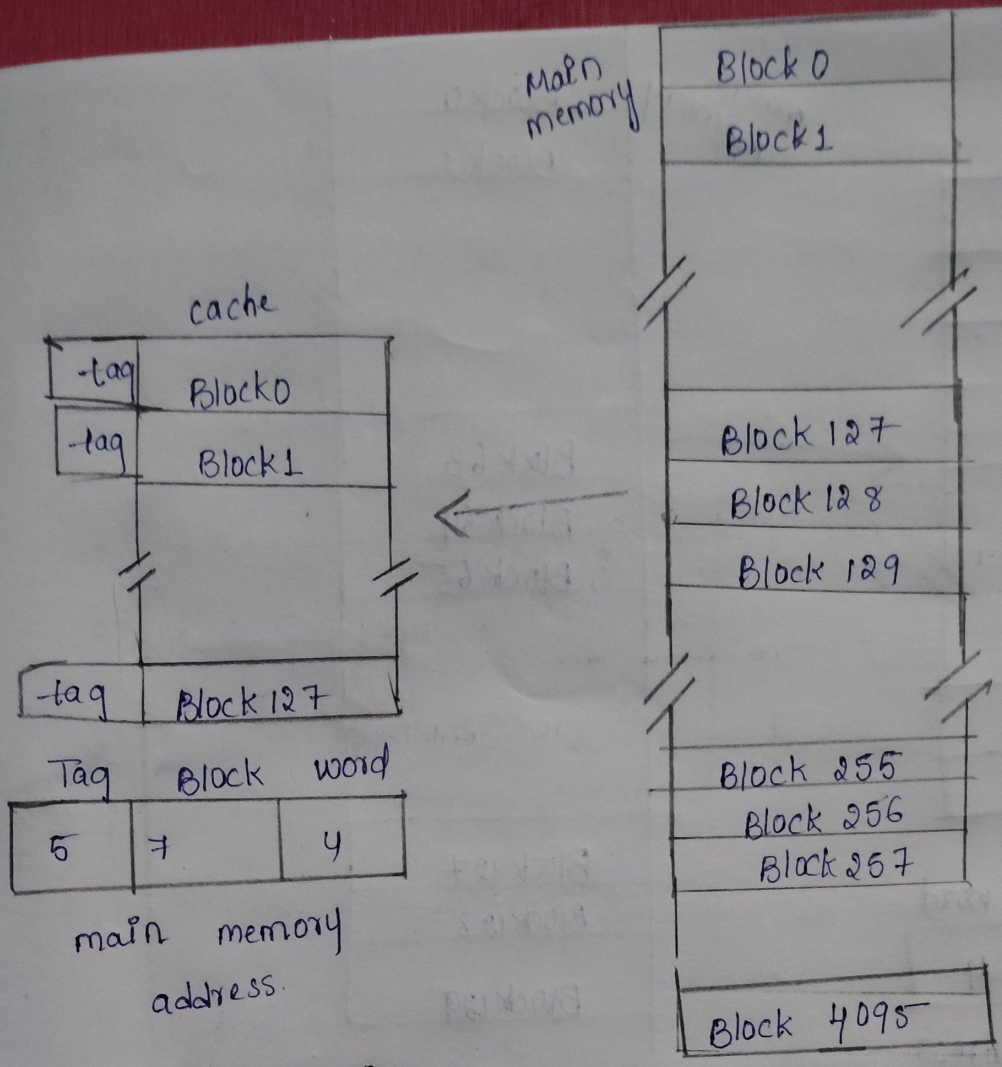
(3) Set-associative mapping.

Direct mapping:↓

→ Block of the main memory maps to j modulo 128 of the cache. 0 maps to 0, 129 maps to 1.

→ memory address is divided into three fields.

⇒ Simple to implement but not very flexible.



Tag	Block	word
5	7	4

main memory address.

Associative mapping:

→ Main memory block can be placed into any cache position

→ memory address is divided into two fields.

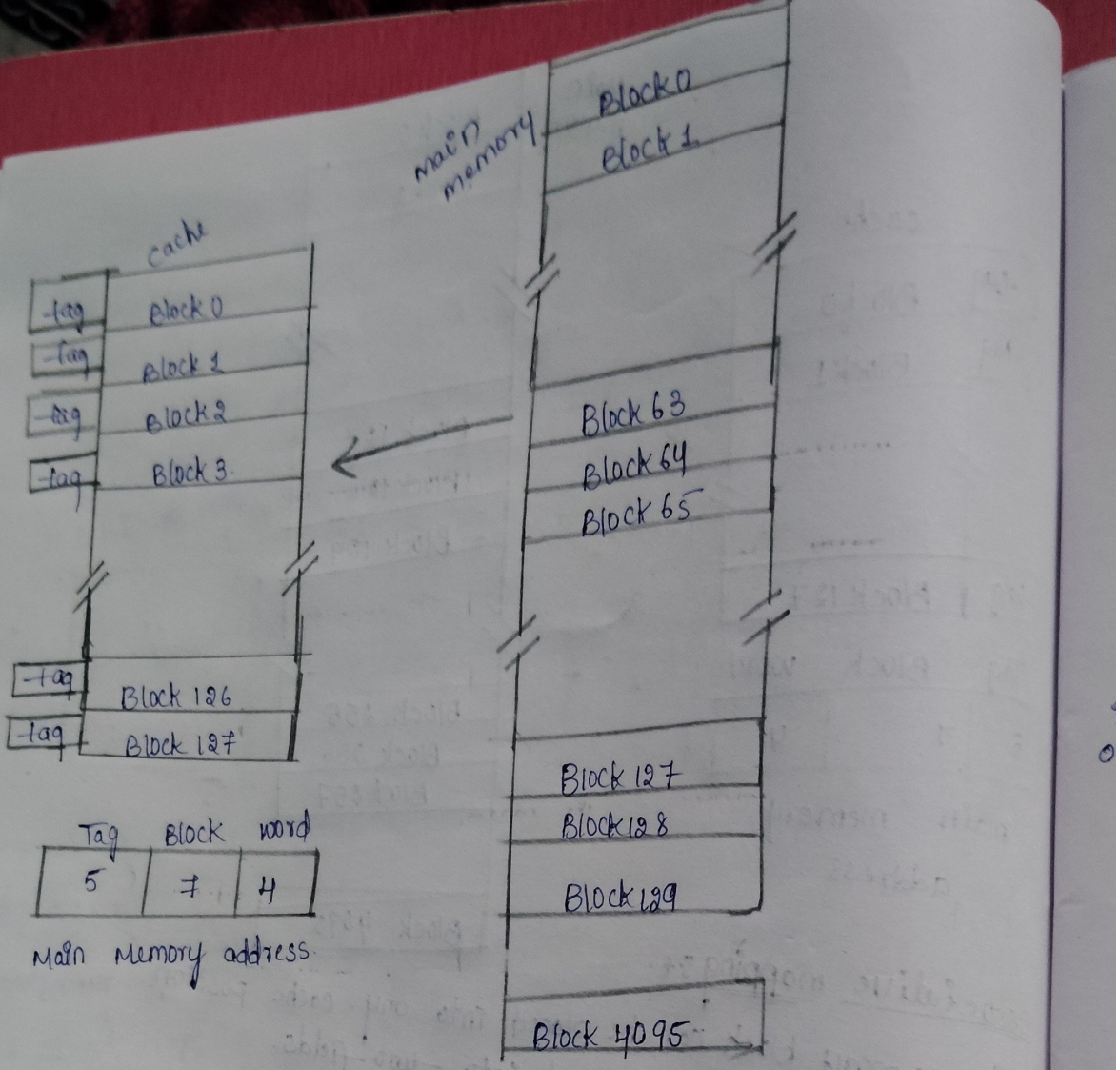
Tag	word
12	4

Main memory address.

set-Associative mapping:

→ Block of cache are grouped into sets

→ memory address is divided into three fields.



(15) Explain interrupts and interrupts hardware. state steps in enabling and disabling interrupts.

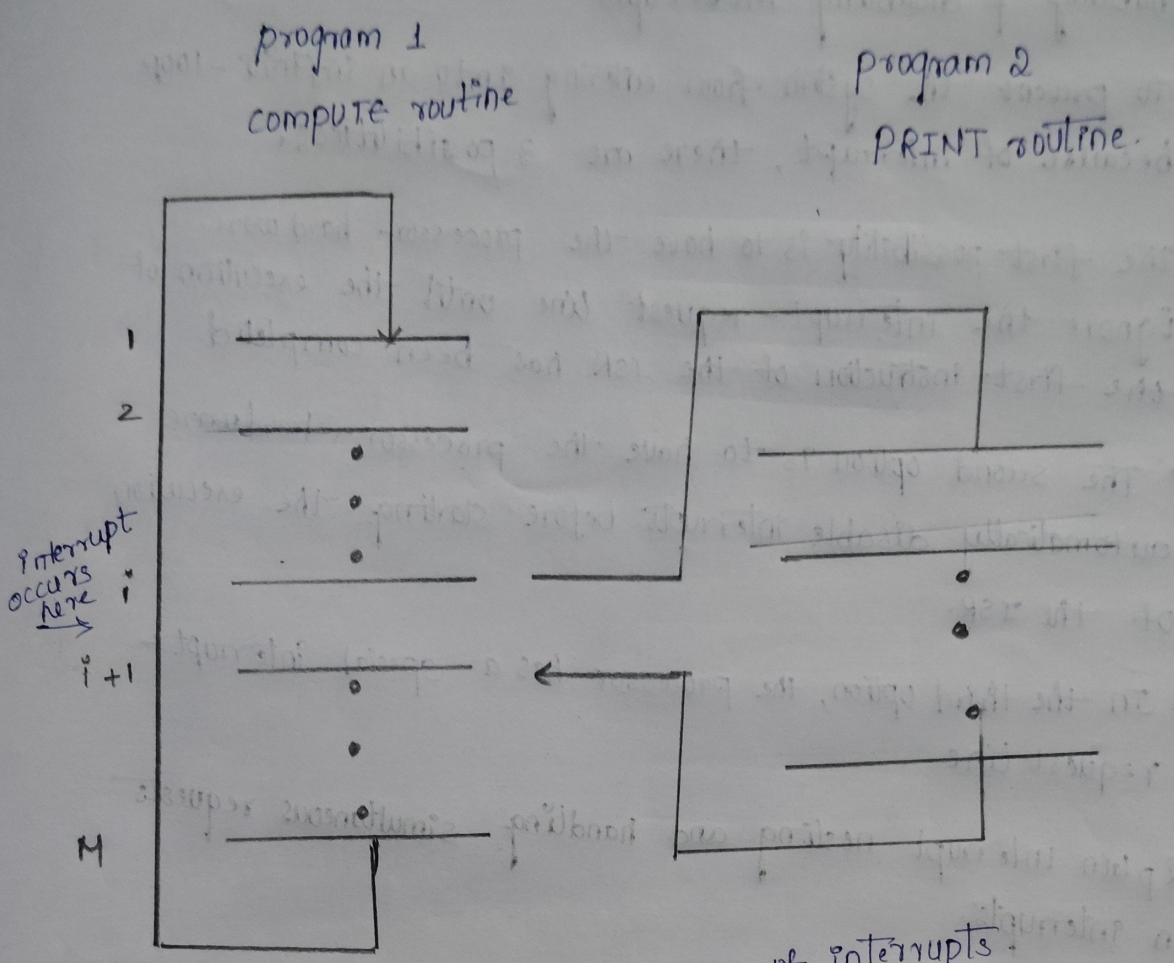
Interrupts:-

→ I/O device initiates the action instead of the processor.

→ This is done by sending a special hardware signal to the processor called as interrupts.

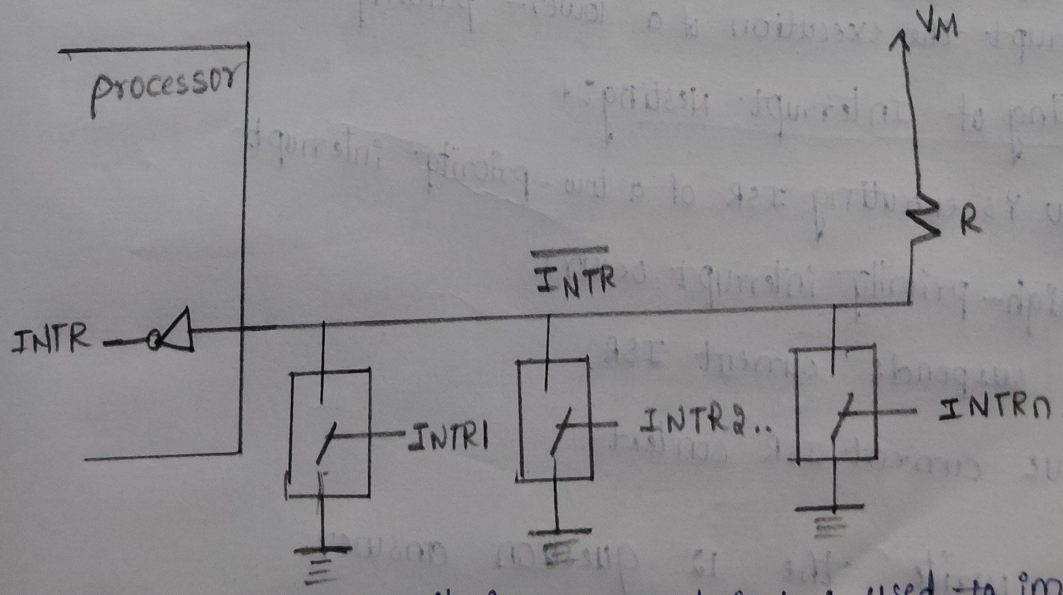
→ The processor can be performing its own task without the need to continuously check the I/O device.

→ Another difference is that an interrupt is a mechanism for coordinating I/O transfers whereas a subroutine is just a linkage of 2 or more function



Transfer of control through the use of interrupts.

Interrupt Hardware:



An equivalent circuit for an open-drain bus used to implement a common interrupt-request line.

Enabling & disabling interrupts.

→ To prevent the system from entering into an infinite-loop because of interrupt, there are 3 possibilities:

1) The first possibility is to have the processor hardware ignore the interrupt-request line until the execution of the first instruction of the ISR has been completed.

2) The second option is to have the processor ~~hardware~~ automatically disable interrupts before starting the execution of the ISR.

3) In the third option, the processor has a special interrupt-request line.

(16) Explain interrupt nesting and handling simultaneous requests in interrupts.

Interrupt Nesting:

Interrupt nesting allows a higher-priority interrupts to interrupt the execution of a lower-priority ISR.

Working of Interrupt Nesting:

→ CPU is executing ISR of a low-priority interrupt
→ A high-priority interrupt occurs
→ CPU suspends current ISR
→ save current ISR context.

And write the 12 question answer.